



# Product datasheet

## Product series: RC series

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## Change history

| Version number | Pages | Revised Content Description                            | Date of revision | Revision Personnel |
|----------------|-------|--------------------------------------------------------|------------------|--------------------|
| V1.0           | -     | First edition enacted                                  | 2024.10.18       | 黄瑜萍                |
| V1.1           | -     | The name C2 should be changed to RC                    | 2024.11.07       | 黄瑜萍                |
| V1.2           | -     | The standard packaging was changed<br>to a flight case | 2024.12.10       | 黄瑜萍                |
| V1.3           | -     | Cabinet Update                                         | 2025.05.28       | 黄瑜萍                |

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# 1.0 Scope of application

This specification is suitable for RC series indoor 500mm\*500mm size standard box, if there are special requirements, it can be customized according to the actual situation

## 2.0 Product Description:

### 2.1 Product Introduction

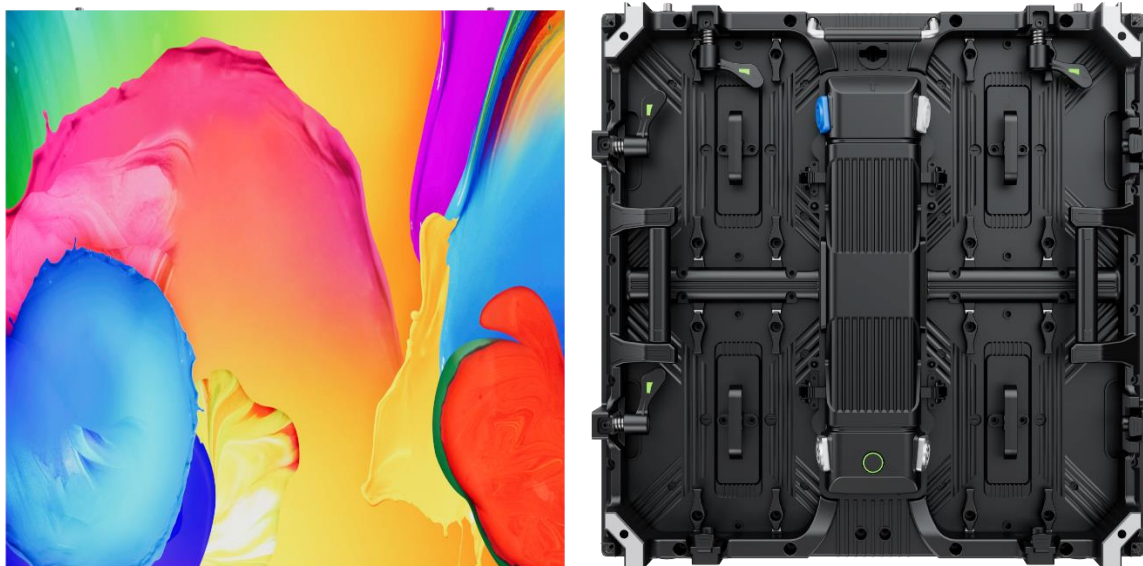
The RC series is a standardized indoor and outdoor rental product developed for the global professional display market. This model is made of die-cast aluminum material, which has the characteristics of light weight, high precision, ultra-thin and good thermal conductivity. The standardized cabinet can be compatible with a variety of spacing, which can be quickly installed indoors, and the module has a variety of characteristics such as anti-dust accumulation and magnetic fixation. It is widely used in indoor concerts, hotel stages, etc.

### 2.2 Features

- a) The positioning structure is used around to ensure the flatness of the screen construction. The box size is 500\*500 standard, and the different sizes of the box support mixed up, down, left and right.
- b) The high-efficiency lock design is adopted between the boxes, and the operation can be completed by one person.
- c) The power supply adopts aviation plug design, which is convenient and stable for connection.

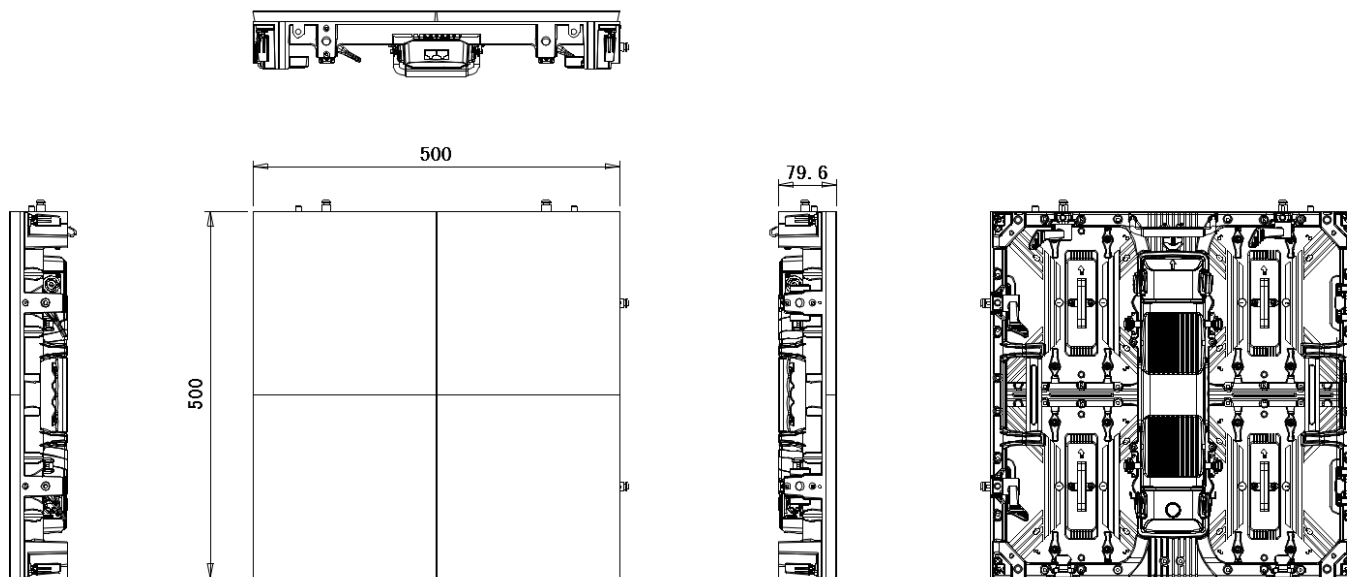
- d) Support internal and external arcs, flexible splicing, curved surface, 90° right angle, dislocation splicing and other diversified splicing methods;
- e) The module adopts hard connection, and the signal is safe and reliable.
- f) The power box adopts an integrated quick-lock structure, which can be quickly disassembled and replaced.
- g) Support post-maintenance, easy and fast installation and maintenance
- h) The box is equipped with a rotating corner protection design, which plays a better role in the transportation and construction process.
- i) Support hoisting. Folding back frame, trolley and other rental accessories supporting design.

## 2.3 Product images



Product diagram

## 2.4 Product size structure drawing



Note: 1. The design and specifications will be upgraded and updated from time to time and will be subject to change without notice.

2. The dimension values (mm) shown in the chart are approximate.

## 3.0 Technical parameters

### 3.1 Mechanical parameters

| project |                                      | parameter         |         |         |         |
|---------|--------------------------------------|-------------------|---------|---------|---------|
| Module  | Pixel pitch (mm)                     | 1.953             | 2.604   | 2.976   | 3.906   |
|         | Pixel composition                    | SMD1515           | SMD1515 | SMD2020 | SMD2020 |
|         | Module resolution (dots)             | 128*128           | 96*96   | 84*84   | 64*64   |
|         | Module size (mm)                     | 250*250           | 250*250 | 250*250 | 250*250 |
| Box     | Cabinet resolution                   | 256*256           | 192*192 | 168*168 | 128*128 |
|         | Pixel density (dots/m <sup>2</sup> ) | 262144            | 147456  | 112896  | 65536   |
|         | Cabinet size (mm)                    | 500*500*79.6      |         |         |         |
|         | Cabinet module composition           | 2 column x 2 row  |         |         |         |
|         | Maintenance methods                  | rear maintenance  |         |         |         |
|         | Cabinet material                     | Die-cast aluminum |         |         |         |
|         | Cabinet flatness                     | ≤0.15mm           |         |         |         |

## 3.2 Optical parameters

| project                              | parameter                              |
|--------------------------------------|----------------------------------------|
| Grayscale (Bit)                      | 14                                     |
| White balance brightness (nit)       | 800                                    |
| contrast                             | 8000: 1                                |
| Color gamut                          | ≥100% NTSC                             |
| Color coordinates                    | standard (0.2953, 0.3047) , adjustable |
| Temperature (K)                      | Standard 8000, 3200~12000 adjustable   |
| Horizontal/Vertical Viewing Angle    | 160°/140°                              |
| Chromaticity uniformity              | ≤0.005                                 |
| Brightness uniformity                | ≥95%                                   |
| Single-point brightness correction   | In the tank                            |
| Single-point chromaticity correction | In the tank                            |

## 3.3 Electrical parameters

| project                                                | parameter          |
|--------------------------------------------------------|--------------------|
| Average power consumption per box (W/box).             | 39                 |
| The maximum power consumption of a single box (W/box). | 115                |
| Average power consumption per square meter (W/m²)      | 157                |
| Maximum power consumption per square meter (W/m²)      | 463                |
| Module power supply requirements                       | DC4.5V             |
| Enclosure power supply requirements                    | AC100~260V/50~60Hz |

## 3.4 System parameters

| project | parameter |
|---------|-----------|
|---------|-----------|

|                             |                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------|
| Frame change frequency (Hz) | 50~60                                                                                |
| Refresh rate (Hz)           | 2880~3840                                                                            |
| Drive mode                  | PWM Constant current drive                                                           |
| Optional dual backup        | Current-sharing power supply                                                         |
| Installation:               | Fixed installation, right-angle installation, internal and external arc installation |

### 3.5 Packaging parameters

| project                       | parameter              |
|-------------------------------|------------------------|
| Net weight of the module (Kg) | 0.62                   |
| Net weight of the box (Kg)    | 6.5 (Modules included) |

Note: The above 3.1~3.5 parameters are the test and calculation parameters of the sample, and the parameters are for reference only because the specific batch configuration may be different;

### 3.6 Environmental requirements

| project                          | parameter               |
|----------------------------------|-------------------------|
| Operating temperature range (°C) | -10~40                  |
| Operating humidity range (RH)    | 10%~65% No condensation |
| Storage temperature range (°C)   | -20~50                  |
| Storage humidity range (RH)      | 10%~65% No condensation |
| Ingress protection (front/rear). | IP30/IP30               |

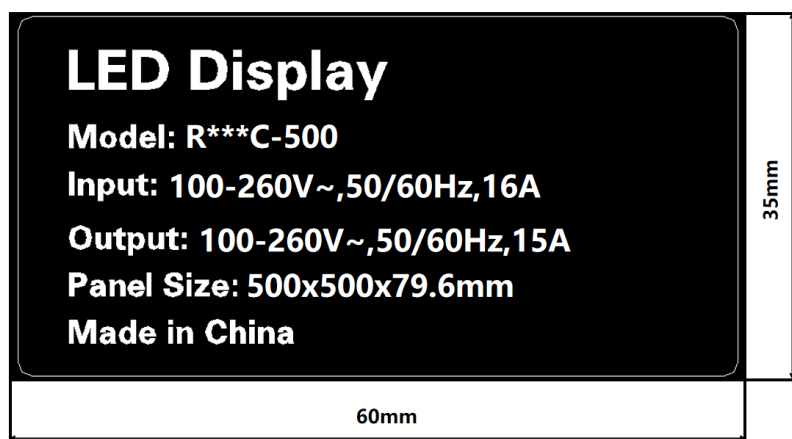
## 4.0 Product interfaces

| legend                                                                              | type            | illustrate                                          |
|-------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------|
|  | Power connector | LED display power interface, maximum current<br>16A |



|                                                                                   |                     |                                                    |
|-----------------------------------------------------------------------------------|---------------------|----------------------------------------------------|
|  | Signal network port | RJ45 network port to transmit video source signals |
|-----------------------------------------------------------------------------------|---------------------|----------------------------------------------------|

## 5.0 Product nameplate

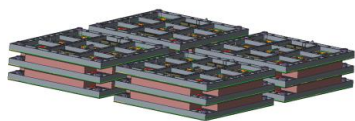


Note: Due to different production batches, the nameplate of this product is optional, if you have special nameplate style or size requirements, please contact us in advance

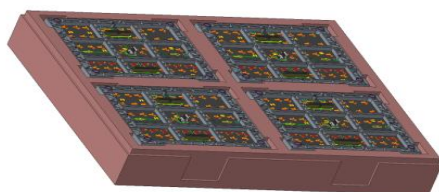
## 6.0 Packaging solutions

Unless the customer specifies its packaging information, it will provide customers with standard flight boxes, and the specific transportation method will vary slightly due to different regions and quantities, and the actual packaging method shall prevail. The packaging method and label information are as follows.

### 6.1 Module packaging solutions



Pearl cotton septum



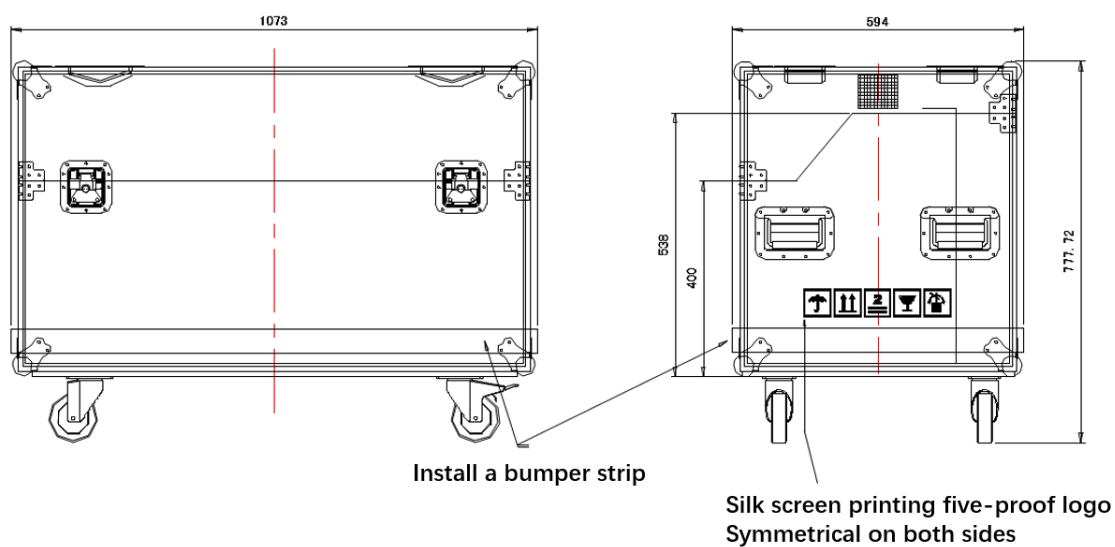
Protected by pearl cotton

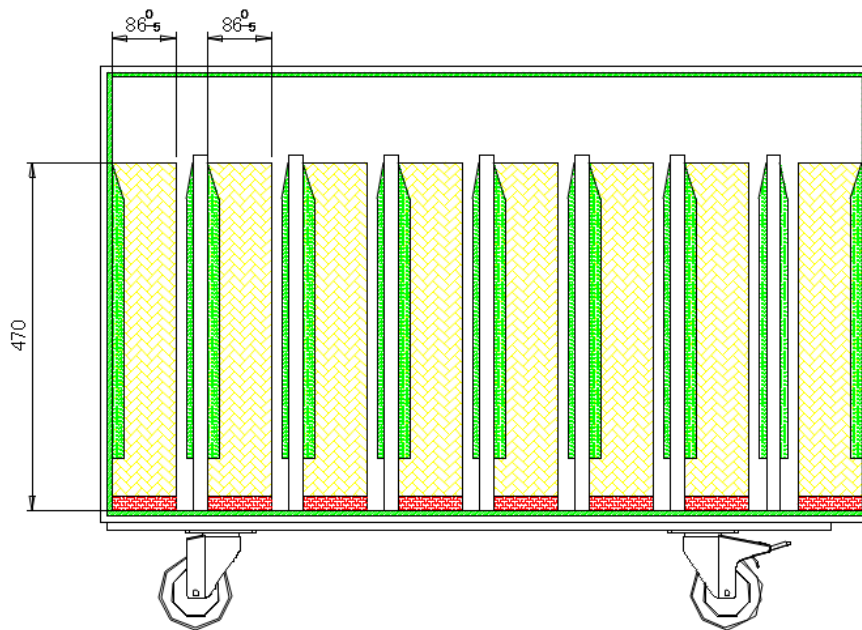


Standard box

## 6.2 Box packaging solutions

**Flight case 1 pack of eight (standard):**





## 7.0 Precautions for product use

| Item                              | Parameter & description                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating environment temperature | <p>Ambient temperature: -10°C~40°C; Other temperature ranges require the addition of temperature control equipment.</p> <p>The temperature of the lamp surface <math>\leq 60^{\circ}\text{C}</math> during the working process, and the temperature exceeds the standard and the temperature control equipment needs to be added;</p> |
| Storage environment temperature   | <p>Ambient temperature: -20°C~50°C; Above 30 °C requires cooling treatment; Exceeded -20°C needs to be heated up;</p>                                                                                                                                                                                                                 |
| Operating environment humidity    | <p>Ambient humidity: 10 ~ 65% RH, humidity exceeding the standard needs to be dehumidified in the use of the environment before normal use;</p>                                                                                                                                                                                       |
| Storage environment               | <p>Ambient humidity: 10~ 65% RH, humidity more than 80%</p>                                                                                                                                                                                                                                                                           |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| humidity                                     | RH needs to be dehumidified;                                                                                                                                                                                                                                                                                                                                                                                                    |
| Overdue storage treatment                    | The product has not been stored for more than one month or power out of the box for more than 48 hours after opening the box, and it takes 6 hours of aging before use before normal use. The aging mode is: full brightness setting 10% 1hrs, full brightness setting 30% 1hrs, full brightness setting 60% 2hrs, full brightness setting 80% 1hrs, full brightness setting 100% 1hrs (brightness gradually increasing aging); |
| dustproof                                    | Ip3X dustproof rating on the back. Should not be exposed to more dust environment, such as more sand and dust outdoors, indoor high dust workshop, etc., need to display the box for special protection;                                                                                                                                                                                                                        |
| Corrosive gas resistant                      | The cabinet should not be exposed to environments where the air contains salt or acid fumes, as this can cause corrosion of electronic components, crystalline leakage, and other issues.                                                                                                                                                                                                                                       |
| Protection against electromagnetic radiation | The module should not be placed in an environment where electromagnetic radiation and radio frequency radiation exceed the intensity of the electric field by 5V/m interference sources.                                                                                                                                                                                                                                        |
| Anti-static                                  | Prevent the damage of electrostatics to electronic devices in humid environments and avoid the damage of leakage to the human body.                                                                                                                                                                                                                                                                                             |
| shockproof                                   | The display should be installed on a robust and reliable installation structure without strong vibration;                                                                                                                                                                                                                                                                                                                       |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stay away from water sources | The box is not waterproofed, and the water conduction can cause the circuit to be short-circuited, resulting in the damage of the circuit components, so it is necessary to stay away from the water source.                                                                                                                                                                                                                                                                                                                                                                                                      |
| personal injury              | The angle and height of the module installation should be suitable, and the sharp corners should be packaged to prevent the damage of the strong outer shell to the human body;                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Security requirements        | It is necessary to ensure the stability of the power supply system, maintain the normal level of frequency, voltage, current harmonics, when a plurality of LED display boxes are spliced, the metal shell should be grounded together with the metal frame installed in the LED display, and the grounding resistance of the overall system of the display should not be greater than $1\Omega$ , and the leakage current should not exceed 5% of the input current. The electrical strength should meet the provisions of SJ/T 1114 1-2017 General Specification 5.7.5 for light-emitting diode (LED) displays; |
| other                        | If the product is an indoor installation product, it should be confirmed that the structure of the product is stable and the safety of the surrounding personnel;                                                                                                                                                                                                                                                                                                                                                                                                                                                 |